Integrated Systems Technology

Laboratory 1: Bulk MOSFET process and electrical characterization

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1 Introduction

Aim of the laboratory: simulate with Synopsys Sentaurus the fabrication process of an n-type bulk MOSFET with nominal gate length $L = 0.18\,\mu\text{m}$ (Sentaurus Process) and then extract its electrical characteristics (Sentaurus Device, Sentaurus Inspect). [TODO: Short description of the workflow and of the tools used: Process, Device, Visual, Inspect.]

2 Process simulation

2.1 Substrate definition and initial mesh

Initial domain, mesh strategy (fine at the surface, coarser in depth) and n-type substrate with $N_D = 10^{15}\,\text{cm}^{-3}$ of phosphorus. [TODO: Comment on the mesh and why it is graded; optional snapshot of Fig. 1.1 equivalent.]

SVISUAL: initial mesh of the substrate

Figure 1: Initial 2D mesh of the substrate.

2.2 Boron implantation (p-well, punch-through stop, V_{th} adjust)

Three boron implants (200 keV, 80 keV, 25 keV) followed by an anneal at 1050 °C for 10 s.

Inspect: B profiles after each step (.plx files)

Figure 2: Boron concentration versus depth after each implant and after the anneal.

[TODO: Zoom in, explain the differences between the profiles (peak position and height versus energy and dose, effect of the anneal), supported by the .tdr snapshots.]

2.3 Gate oxide growth

Dry oxidation at 850 °C for 10 min.

SVISUAL: oxide on the substrate, with ruler measurement

Figure 3: Gate oxide after thermal oxidation.

[TODO: Oxide thickness measured with the ruler: $t_{ox} = \ldots$ nm.]

2.4 Gate creation and polysilicon reoxidation

Polysilicon deposition (0.18 µm), patterning with `gate_mask` (half transistor, mirrored at the end), anisotropic etch of polysilicon and oxide, reoxidation at 900 °C for 10 min. [TODO: Snapshots of NMOS1, NMOS2, NMOS3 and a brief comment on each etch step.]

2.5 LDD implantation

Arsenic implant ($4 \times 10^{14}\,\mathrm{cm}^{-2}$, 10 keV) and short anneal. Refinement box defined before the implant. [TODO: Snapshot of the activated As profile (AsActive) with a suitable range.]

2.6 HALO implantation

Four boron implants with tilt 30° and rotations 0°, 90°, 180°, 270°, followed by an RTA of 5 s at 1050 °C.

Halo 1

(a) Halo 1 (rotation 0°).

Halo 2

(b) Halo 2 (rotation 90°).

Halo 3

(c) Halo 3 (rotation 180°).

Halo 4

(d) Halo 4 (rotation 270°).

Figure 4: Structure after each HALO implantation.

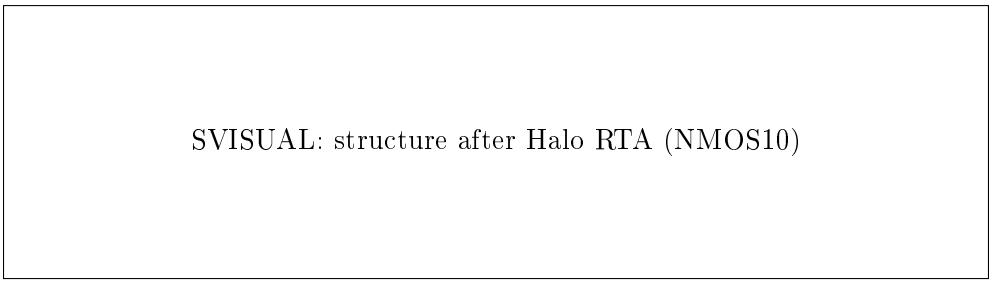


Figure 5: Boron profile after the HALO RTA.

[TODO: Comment the effect of each of the four implants and of the RTA (symmetry, reduction of charge sharing / punch-through).]

2.7 Spacers formation

Isotropic nitride deposition (60 nm), anisotropic etch with a small isotropic overetch, removal of the thin oxide. [TODO: Snapshots of NMOS11, NMOS12, NMOS13 and a brief comment.]

2.8 Source/drain implantation

Refinement box in the S/D region (coordinates of Fig. 1.11 of the handout, uniform spacing of 10 nm along x and 50 nm along y).

```
% paste here the refinebox command lines actually used
refinebox Silicon min= {...} max= {...} xrefine= {...} yrefine= {...} add
grid remesh
```

[TODO: Comment the command lines (meaning of min, max, xrefine, yrefine, grid remesh).]

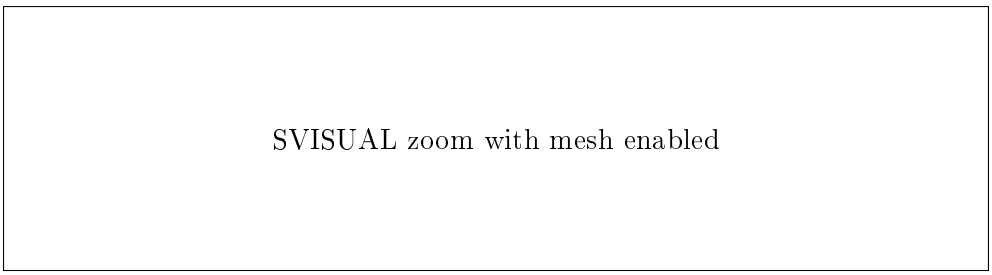


Figure 6: Local mesh refinement in the S/D region.

Then Arsenic implant ($5 \times 10^{15}\,\mathrm{cm}^{-2}$, 40 keV, tilt 7°) and final RTA. [TODO: Snapshot of NMOS14 and NMOS15.]

2.9 Contacts and final structure

Aluminium deposition and patterning (S/D pads and gate pad), mirroring of the half structure (`transform reflect left`), definition of the contacts (substrate, gate, source, drain). [TODO: Snapshots of NMOS16–NMOS18, of the final structure and of the presimulation structure.]

3 Electrical simulation

3.1 Simulation setup

Brief description of the `sdevice_des.cmd` file: sections File, Electrode, Physics (doping-dependent mobility, high-field saturation, SRH), Math and Solve (ramp of V_{DS} up to 2 V, then ramp of

V_{GS} up to 2 V). [TODO: Add the two additional quasistationary blocks for $V_{DS} = 0.2$ V and the `Load` / `NewCurrentPrefix` lines used.]

3.2 Transcharacteristic

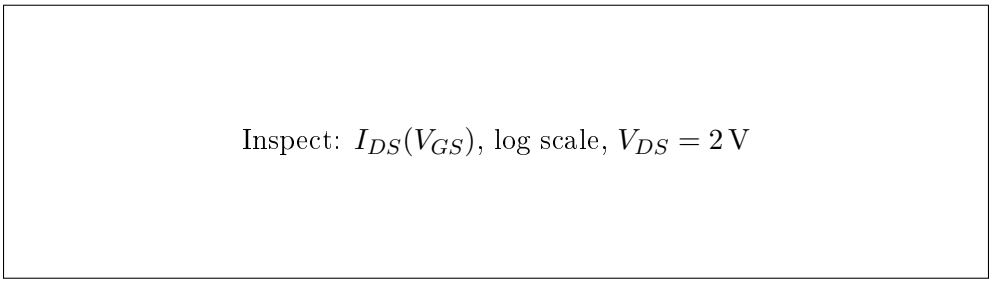


Figure 7: $I_{DS}(V_{GS})$ characteristic in semi-logarithmic scale at $V_{DS} = 2$ V.

3.3 I_{OFF} and I_{ON}

Table 1: Off and on currents ($V_{DS} = 2$ V, per µm of width).

Quantity	Definition	Value
I_{OFF}	I_{DS} at $V_{GS} = 0$ V	[TODO: value] A µm^{-1}
I_{ON}	I_{DS} at $V_{GS} = 2$ V	[TODO: value] A µm^{-1}

3.4 Threshold voltage and subthreshold slope

V_{th} from the second-derivative method and SS from the maximum slope of $\log_{10} I_{DS}$:

$$V_{th} = V_{GS}\Big|_{\max \frac{d^2 I_{DS}}{dV_{GS}^2}}, \qquad SS = \left[\max \frac{d \log_{10} I_{DS}}{dV_{GS}}\right]^{-1} \tag{1}$$

Table 2: Figures of merit extracted with Sentaurus Inspect.

	$V_{DS} = 0.2$ V	$V_{DS} = 2$ V
V_{th} (V)	[TODO:]	[TODO:]
SS (mV/dec)	[TODO:]	[TODO:]

3.5 DIBL

$$\mathrm{DIBL} = \frac{V_{th}(V_{DS,low}) - V_{th}(V_{DS,high})}{V_{DS,high} - V_{DS,low}} \quad [\mathrm{mV/V}] \tag{2}$$

[TODO: Plot the two transcharacteristics on the same graph, compute DIBL and comment the value.]

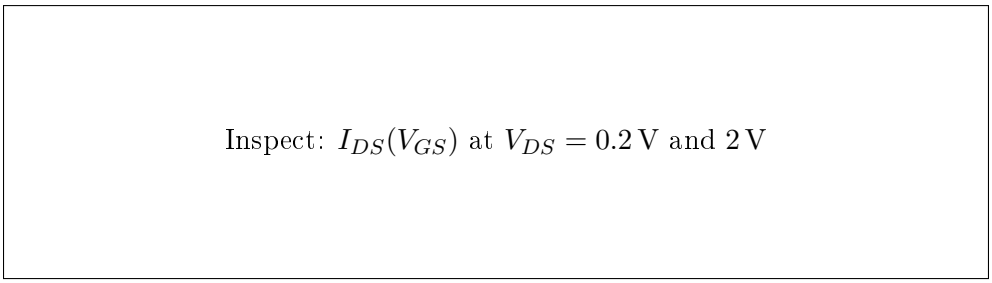


Figure 8: Transcharacteristics at low and high drain bias.

4 Conclusions

[TODO: Summarize the main results (V_{th}, SS, DIBL, I_{ON}/I_{OFF}), comment on the role of LDD, HALO and punch-through implants, and on short-channel effects of the 0.18 µm device.]